

Large-scale experimental data, corpus examples, and crosslinguistic data

Deniz Özyıldız & Ciyang Qing

University of Konstanz & University of Edinburgh

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Day 5

Clause-embedding predicates

- ▶ **Responsive:** ✓declarative, ✓interrogative

E.g., *know*, *be certain*

- (1) Alice knows ✓that/✓whether Bob is in Amsterdam

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(complications to be discussed)

Potential semantic predictors

Big picture questions:

- ▶ What semantic properties of a clause-embedding predicate may influence its compatibility with different clause types?
- ▶ How and why?

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- ▶ Neg-raising (Zuber 1982; Theiler et al., 2019)
- ▶ Preferentiality (Uegaki & Sudo, 2019)

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Limitation: only a small set of paradigmatic examples

- ▶ How robust are these generalization?

The larger empirical landscape

Large datasets at the scale of the lexicon

- ▶ MegaAcceptability (White & Rawlins, 2016, 2020)
- ▶ MegaVeridicality (White & Rawlins, 2018)
- ▶ MegaNegRaising (An & White, 2020)

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White (2021) uses these quantitative datasets, together with corpus examples, to challenge several existing empirical generalizations in the literature:

“I have presented evidence from both large-scale experimental data and examples attested in corpora that **these generalizations are, if not outright falsified, so frail that they should be jettisoned.**” (p. 16)

High-level messages

How should we use large-scale, quantitative experimental data to assess empirical generalizations and formal analyses in semantics?

- ▶ Standard: testing certain summary statistics
- ▶ In-depth qualitative investigation on individual items is (also) important

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- ▶ Constructive theory building/revision by critically assessing assumptions in the derivation

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Crosslinguistic investigations are fruitful

- ▶ You can contribute!

Neg-raising and responsivity

Zuber (1982): **neg-raising predicates are anti-rogative**

(3) Alice thinks ✓that/*whether Bob is in Amsterdam

(4) Alice wants (*whether) to visit Amsterdam

More examples: *believe, seem, be likely*

Neg-raising inference

The negation of a neg-raising predicate is interpreted as if the negation were in the embedded clause

Neg-raisers, e.g., *think*, *believe*:

- (5) Alice didn't think/believe that Bob was here
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- (6) Alice didn't say that Bob was here
 $\not\approx$ Alice said that Bob was not here
- (7) Alice didn't know that Bob was here
 $\not\approx$ Alice knew that Bob was not here

I will call this the **inferential diagnostic** of neg-raising

Quantitative measures

MegaNegRaising (An & White 2020)

- (8) If I were to say *I don't think that a particular thing happened*, how likely is it that I actually mean that *I think that that thing didn't happen*?

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- (9) Someone {thought, knew, ...} that/whether something happened.
- ▶ Test sentences are semantically bleached: intended to minimize plausibility effects
 - ▶ Quantitative scores obtained after normalization

Quantitative prediction of NR

Zuber's (1982) empirical generalization NR:
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- ▶ White (2021): NR predicts that neg-raising and responsivity scores are (generally) negatively correlated

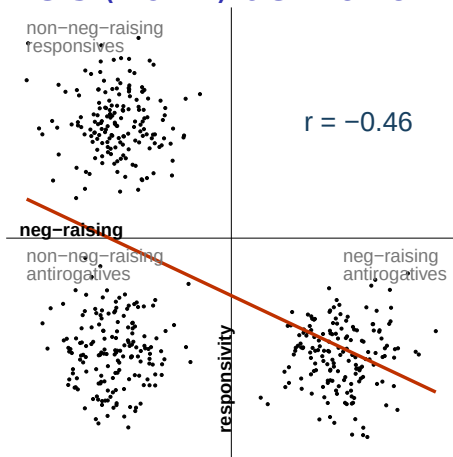
Quantitative prediction of NR

Zuber's (1982) empirical generalization NR:

Neg-raising predicates are anti-rogative/not responsive

- ▶ White (2021): NR predicts that neg-raising and responsivity scores are (generally) negatively correlated
- ▶ This initially sounds plausible, and White (2021) provides a derivation assuming discrete categories of neg-raising and responsive predicates

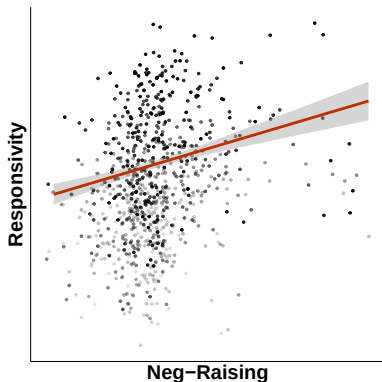
White's (2021) derivation of quantitative NR



NR: Neg-raising predicates are not responsive

- ▶ The top-right corner must be empty (modulo noise)
- ▶ White's (2021) simulation: $r > 0.09$ highly unlikely

Observed correlation



$$r = 0.22 \text{ (95\%CI = [0.14, 0.30])}$$

White (2021): “This correlation violates the predictions of NR , even under the worst-case simulation assumptions discussed above, suggesting that NR is false as a generalization about the lexicon as a whole”

Complication: subject + tense

MegaAcceptability (White & Rawlins, 2016, 2020)

(10) Someone {thought, knew, ...} that/whether something happened.

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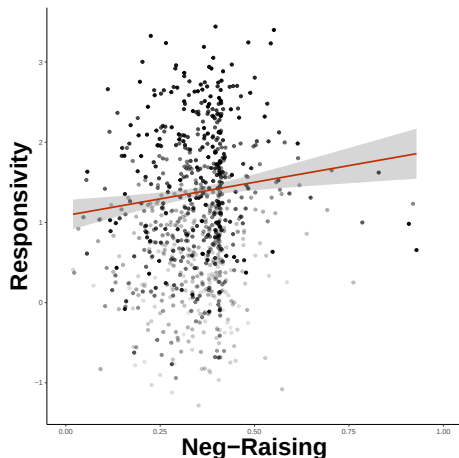
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MegaNegRaising (An & White 2020)

(11) If I were to say *I don't think that a particular thing happened*, how likely is it that I actually mean that *I think that that thing didn't happen*?

- ▶ {1sg, 3sg} + {present, past}
- ▶ Maximum as neg-raising score

Overall correlation: 3sg+past



$r = 0.11$ (95%CI = [0.03, 0.18])

(originally $r = 0.22$)

Limitation of statistical testing

How should we use large-scale, quantitative experimental data to assess generalizations and formal analyses in semantics?

- ▶ Standard: testing certain summary statistics

Limitation:

- ▶ Test result can only reject a hypothesis: no information about where and why it fails

Further exploring the data

NR: Neg-raising predicates are not responsive

- ▶ Direct falsification: predicates that are neg-raising and responsive

Further exploring the data

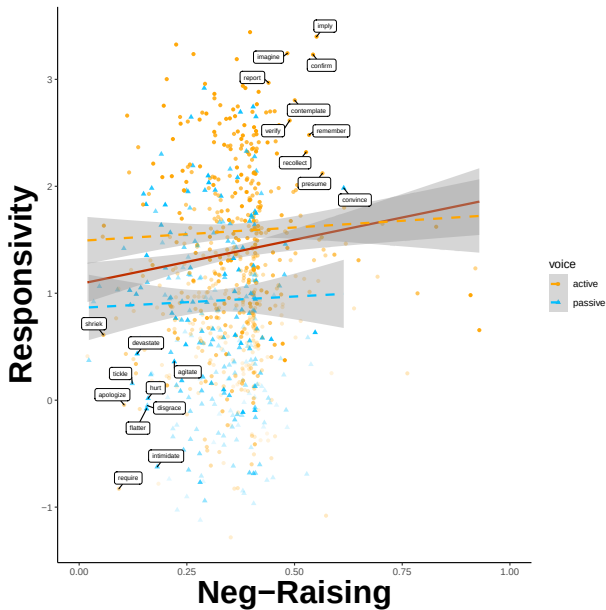
NR: Neg-raising predicates are not responsive

- ▶ Direct falsification: predicates that are neg-raising and responsive

How to find such counterexamples efficiently?

- ▶ Challenge: noisy data, no clear thresholds
- ▶ But: high neg-raising and responsivity scores
⇒ large contribution to the positive correlation
- ▶ Heuristic: start with top contributors

Top 20 contributors



10 potential counterexamples

10 predicates in the top-right corner

- ▶ Class A: *confirm, report, verify, remember, recollect, imply, be convinced*
- ▶ Class B: *imagine, presume, contemplate*

They are all compatible with interrogative complements

- ▶ They are potential counterexamples to NR
- ▶ But do they really falsify it?

Complication 1: What is neg-raising?

Descriptively, it is a phenomenon where it looks as if the negation in an embedded clause were raised to negate the predicate in a higher clause.

- ▶ **Inferential diagnostic:** $\neg V\phi$ implies $V\neg\phi$
- ▶ **Strong-NPI licensing:** $\neg V\phi[\checkmark \in \text{Strong-NPI}]$

Neg-raising: strong NPI licensing

The negation of a neg-raising predicate licenses strong NPIs in the embedded clause

- (12) Alice didn't think/believe that Bob had been here
 in years
 ≈ Alice thought/believed that Bob had not been
 here in years

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≈ Alice thought/believed that Bob had not been here in years

(13)*Alice didn't say that Bob had been here in years

Cf. Alice said that Bob had not been here in years

(14)*Alice didn't know that Bob had been here in years

Cf. Alice knew that Bob had not here in years

Pragmatic account of neg-raising inference

If we only care about the neg-raising inference, note that it will go through whenever the context happens to support an **Excluded Middle (EM)** assumption:

$$V\phi \vee V\neg\phi$$

- If $\neg V\phi$, the second disjunct above must be true

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- ▶ If $\neg V\phi$, the second disjunct above must be true
- ▶ This is essentially Bartsch's (1973) pure pragmatic account of neg-raising, and presumably should be the null hypothesis (Jacobson 2020)
- ▶ However, Horn (1978, 1989) observes that non-neg-raisers fail to license strong NPIs in contexts with explicit EM premises

Horn's example with *say*

- (15) (Background: John is Mary's best friend.
Someone else claimed that Mary had been here.)

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 and he told me in no uncertain words.
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- (15) (Background: John is Mary's best friend.
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 He didn't say she's been here (*in weeks).

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He didn't say she's been here (*in weeks).
(cf. he said she hasn't been here in weeks)

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 and he told me in no uncertain words.
 He didn't say she's been here (**in weeks*).
 (cf. he said she hasn't been here in weeks)

Challenging for a pure pragmatic account to explain the contrast between *say* and *think/believe* in terms of strong-NPI licensing

- ▶ A “neg-raising predicate” is not just a predicate that triggers the neg-raising inference

Which diagnostic of neg-raising is intended?

Zuber's (1982) formal analysis: the following 3 properties are jointly inconsistent

- ▶ $\neg O(\text{that } S)$ implies $O(\text{that } \neg S)$ (neg-raising)
- ▶ $O(\text{that } p)$ and $O(\text{that } \neg p)$ both imply $O(\text{whether } p)$ (Prop 1)
- ▶ $\neg O(\text{that } p)$ implies $\neg O(\text{whether } p)$ (Prop 2)

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The inferential diagnostic is (presumably) appropriate for Zuber's (1982) intended notion of neg-raising

- ▶ But note that the experimental data do not contain measures of Prop 1 and Prop 2!

Theiler et al.'s (2019) derivation of NR

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- ▶ Semantic assumption: *believe* is a neg-raising counterpart of *be certain*
- ▶ The at-issue content of $B(\llbracket \text{whether } \varphi \rrbracket)$ is equivalent to $B(\varphi) \vee B(\neg\varphi)$ (cf. *be certain*)
- ▶ *Believe* carries an Excluded Middle Presupposition (EMP) $B(\varphi) \vee B(\neg\varphi)$, which is what allows for the neg-raising inference
- ▶ Note that the at-issue content of $B(\llbracket \text{whether } \varphi \rrbracket)$ is the same as its EMP, i.e., *believe whether* φ does not contribute anything more than what it already presupposes.
- ▶ Theiler et al.'s (2019) takes this triviality to be the reason for the antirogativity of *believe*

Theiler et al.'s (2019) notion of neg-raising

Theiler et al. (2019, fn. 13):

“*Be true/false* aren’t categorized as neg-raising here, although they do license neg-raising inferences. This is because, as illustrated in (i), negated *be true/false* don’t license strict NPIs, unlike predicates like *think* and *believe* [. . .]

(i) a. *It isn’t true that Mary will leave until June.

b. John doesn’t think that Mary will leave until June. ”

Class A examples are not “real” neg-raisers

Class A: *confirm, report, verify, remember, recollect, imply, be convinced*

- (16) *John didn't
confirm/report/verify/remember/recollect that
Mary had been to Amsterdam in years
- (17) ??John didn't imply that Mary had been to
Amsterdam in years
- (18) ?John wasn't convinced that Mary had been to
Amsterdam in years
≠ John was convinced that Mary hadn't been to
Amsterdam in years

Such predicates are not neg-raisers in the sense intended by Theiler et al. (2019) and do not falsify NR.

Class B apparent counterexamples

Class B: *imagine, presume, contemplate*

- (19) John didn't imagine/presume that Mary had been to Amsterdam in years

According to Green (1974), *contemplate* is neg-raising in the sense that expresses intentions

- (20) I'm not contemplating leaving town until after Easter.
 ≈ I'm contemplating not leaving town until after Easter.

Complication 2: polysemy

Horn (1978) citing Green (1974)

- ▶ *Imagine* is not neg-raising in the literal *form a mental image* sense
- ▶ *Contemplate* is not neg-raising in the *ponder* sense

Similarly, *presume* is not neg-raising in the literal *make a presumption* sense

This suggests that the neg-raising and the question-embedding senses of these predicates are actually different

- ▶ If so, they would not be real counterexamples to NR

Availability of non-habitual readings

- (21) I don't imagine that Mary has been to Amsterdam.
- ▶ Habitual reading possible
 - ▶ Non-habitual reading more natural/salient
≈ I am of the opinion that Mary hasn't been to Amsterdam
- (22) I don't imagine which friends have been to Amsterdam.
- ▶ Habitual reading only
 - ▶ Non-habitual reading impossible

Coordination test

Alice gave a keynote talk at the 2023 annual conference and raised 10 open questions for the field.

- (23) She didn't imagine that these questions would be solved until 2030.
- (24) She didn't imagine which one would have the biggest impact
- (25) ??She didn't imagine that these questions would be solved until 2030 or which one would have the biggest impact

Coordination test: *contemplate*

The committee did not promote John because ...

- (26) they did not contemplate that he had made significant contributions to the field.
 - ▶ Neg-raising reading possible
 - ▶ Non-neg-raising *take into account* reading possible
- (27) they did not contemplate whether his research had broader impacts.
- (28) they did not contemplate that he had made significant contributions to the field or whether his research had broader impacts.
 - ▶ Neg-raising reading impossible
 - ▶ Non-neg-raising *take into account* reading only

Coordination test: *consider*

The committee did not promote John because ...

- (29) they did not consider that he had made significant contributions to the field.
 - ▶ Neg-raising reading possible
 - ▶ Non-neg-raising *take into account* reading possible
- (30) they did not consider whether his research had broader impacts.
- (31) they did not consider that he had made significant contributions to the field or whether his research had broader impacts.
 - ▶ Neg-raising reading impossible
 - ▶ Non-neg-raising *take into account* reading only

Interim summary

In-depth qualitative investigation on individual items reveals confounds

- ▶ Class A predicates license neg-raising inferences without being “real” neg-raisers in the sense intended by Theiler et al. (2019)
- ▶ Class B predicates are polysemous between neg-raising and non-neg-raising senses, but only the non-neg-raising ones are compatible with interrogative complements

No genuine counterexamples to NR

Preview

White (2021) provides corpus examples where *think* and *believe* take interrogative complements

- ▶ We have discussed *think* and will focus on *believe* today

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- ▶ We have discussed *think* and will focus on *believe* today

Even though such corpus examples technically falsify NR as a descriptive generalization, they are in fact compatible with the formal analyses

- ▶ The formal analyses already suggest how NR should be qualified
- ▶ Independently motivated additional mechanisms are needed for such examples.

White's (2021) claim

“The aim of this paper is to show that, not only are these generalizations (and thus their derivations) nonviable when evaluated across a broad swath of the English verb lexicon; even the inference from the contrast in (1) to the antirogativity of *think*, *believe*, *hope*, and *fear* is unlicensed based on an assessment of the broader empirical landscape. And, I argue, because these predicates’ (purported) antirogativity forms the core data for all of these generalizations, the generalizations are left so frail without these data that they should be jettisoned altogether.” (p. 2)

Zuber's (1982) qualification

'It seems important to point out in conclusion that what I have shown is **not that whether-complementizers are absolutely impossible with neg-transportation verbs**. I have shown that **they are impossible only with their "main" meaning**, the one given by the definition of negtransportation verb and by the properties Prop 1 and Prop 2. In particular they must be related to the that-complementizer in the way shown above. Consequently it may happen that in some languages one can find, in some construction, *+to think whether* or its equivalent. **My claim is however that in this case the meaning of the verb will be different from the one stipulated by the properties here discussed.**' (p. 436)

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Crucially, *believe whether φ* can be fine if either

- ▶ the function-argument structure does not involve $B(\llbracket \text{whether } \varphi \rrbracket)$, or
- ▶ the at-issue content $B(\llbracket \text{whether } \varphi \rrbracket)$ is not equivalent to its EMP

White's counterexample 1

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How is the meaning derived compositionally?

- ▶ If *believe* simply takes the denotation of its interrogative complement as its semantic argument, we would end up with only 1 alternative, i.e., $B(\{p, \neg p\})$

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- ▶ If *believe* simply takes the denotation of its interrogative complement as its semantic argument, we would end up with only 1 alternative, i.e., $B(\{p, \neg p\})$
- ▶ But *torn between* requires two alternatives!

Generality of the pattern

Can it be that *believe* has a special semantics such that $B(\{p, \neg p\})$ ends up generating two alternatives?

- Unattractive, because the pattern is quite general

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(34) I don't recognise it but it looks so much like an original part of the launch sequence that I'm **torn between saying whether** it was shot by Pod 4 Films, or AP Films

(35) Have you ever felt **torn between knowing whether** you're a J or P personality type?

(36) I am often **torn between deciding whether** I should give money or time to help a deserving cause.

Additional mechanism: alternative percolation

Our proposal: the alternatives in the interrogative complement simply “percolate up” and compose with *believe* in a point-wise fashion.

- ▶ $\llbracket \text{believing whether or not } p \rrbracket = B \circ \{p, \neg p\} = \{Bp, B\neg p\}$

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- ▶ $\llbracket \text{believing whether or not } p \rrbracket = B \circ \{p, \neg p\} = \{Bp, B\neg p\}$
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Crucially, under this analysis, *B* in fact does not take its interrogative complement as its semantic argument

White's counterexample 2

- (37) We can choose to **believe whether** the word of God is true[...]or not.

White's counterexample 2

- (37) We can choose to **believe whether** the word of God is true[...]or not.

Similarly, alternative percolation would provide the two alternatives for *choose*:

- ▶ *believe that the word of God is true*
- ▶ *believe that the word of God is not true*

White's counterexample 3

- (38) [...] I didn't believe the Bible growing up, I wasn't a Christian growing up, I struggled to **believe whether** I could trust the Scriptures[...]

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- (38) [...]I didn't believe the Bible growing up, I wasn't a Christian growing up, I struggled to **believe whether** I could trust the Scriptures[...]

A salient interpretation is *I struggled to believe that I could trust the Scriptures*

- Theiler et al. (2019) does not predict unacceptability in this case, because the at issue content of $B(\llbracket \text{whether } \varphi \rrbracket)$ is $B\varphi$, not equivalent to its EMP $B\varphi \vee B\neg\varphi$

White's counterexample 4 interpretation I

- (39) Richard Stavin, a former veteran federal prosecutor[...]declared[...] “[...]I believe it was an organized, orchestrated effort on the part of certain individuals within Washington, D.C. to keep a hands-off policy towards MCA[...]”
Believing whether certain individuals within Washington D.C. had an MCA policy is not the same as proving there was such a policy.

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Let ϕ be *certain individuals within Washington D.C. had an MCA policy*. Alternative percolation would generate an interpretation equivalent to an unconditional:

- Believing that ϕ or believing that $\neg\phi$, it is not the same as proving that there was such a policy.

White's counterexample 4: interpretation II

However, perhaps a more plausible interpretation is *believing that certain individuals within Washington D.C. had an MCA policy is not the same as proving there was such a policy*

- ▶ That is, *believing whether φ somehow ends up meaning believing that φ*

Summary

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Summary

White (2021) provides corpus examples where *believe* takes interrogative complements

- ▶ While these are counterexamples to NR , they can be explained away by other independent mechanisms, e.g., alternative percolation, and therefore are compatible with Theiler et al.'s (2019) formal semantics
- ▶ Of course, there are many open issues
- ▶ But these are problems for everybody: it is not enough to just say *believe* can take interrogative complements, without providing an analysis of the meanings of the counterexamples

Non-veridical preferential predicates

Uegaki & Sudo (2019): Non-veridical preferential predicates such as *hope* are incompatible with interrogative complements (NVP)

- (40) a. Al likes/is surprised which students you invited. (Veridical)
- b. *Al hopes/wishes which students you invited. (Non-veridical)

Uegaki & Sudo's (2019) derivation of NVP

- ▶ The at-issue content of *A hopes Q* is
 $\exists p \in Q. \mu_{\mathbf{hope}}(p) > \theta_C.$

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- ▶ Note that in this case the at-issue content of *hope Q* is the same as its TSP
- ▶ Uegaki & Sudo (2019) take this triviality to be the reason for the antirogativity of *hope*

Ways to avoid triviality

A non-veridical preferential predicate v combining with a question Q would not be trivial if either one holds:

- ▶ The at-issue content is not identical to the TSP, i.e., it does not mean $\exists p \in Q. \mu_v(p) > \theta_C$.
- ▶ V does not have a TSP to begin with!

Hope: White's (2021) counterexamples

- (41) a. I was hoping whether you are able to guide me[...]

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- a. I was hoping whether you are able to guide me[...]
 - b. I have done a quite a bit of research on using a Limited Co but was hoping whether someone with more experience could confirm my understanding of a few points[...]

Hope: White's (2021) counterexamples

- (41)
- a. I was hoping whether you are able to guide me[...]
 - b. I have done a quite a bit of research on using a Limited Co but was hoping whether someone with more experience could confirm my understanding of a few points[...]
 - c. This Trump/Carson boom really has people like Bush, Walker, Rubio, and others wondering and hoping whether history will repeat itself and whether Republicans will return back to focusing on the establishment choices but it's all about outsider candidates right now.

Hope whether $p \approx$ hope that p

In all these cases, *hope whether p* gets interpreted as *hope that p*

- ▶ No attested examples of *hope whether p* being interpreted as *hope that not p*

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In all these cases, *hope whether p* gets interpreted as *hope that p*

- ▶ No attested examples of *hope whether p* being interpreted as *hope that not p*
- ▶ *Hope whether p* and *hope whether $\bar{p}$* are interpreted differently

(42) I was hoping whether you would be in town.
 $\neq$ I was hoping whether you would be out of town.

The at-issue content of *hope whether p* is *hope that p* , which is stronger than the TSP: No triviality!

Fear: White's (2021) counterexamples

- (43) a. Interstellar space is so vast that there is no need to fear whether stars in the Andromeda galaxy will accidentally slam into the Sun.
 (≈ fear that stars will slam into the Sun)

Fear: White's (2021) counterexamples

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($\approx$ fear that I will **not**)

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 - b. I fear whether I'll have use of my arms/hands by age 55 or 60.
($\approx$ fear that I will **not**)
 - c. I know parents who seriously fear whether their children will ever hold a meaningful job.
($\approx$ fear that their children will **n**ever . . .)

Fear whether $p \approx$ fear that $p \vee$ fear that not p

Evaluatively negative predicates: no TSP

Uegaki & Sudo's (2019) Threshold Significance
Presupposition:

► $\exists p \in C. \mu_V(p) > \theta_C$

(44) I don't like that Bo won the game.

↗ There exists someone x s.t. I would have liked
it if x won the game)

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 $\rightsquigarrow$ There exists someone x s.t. I would have liked
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But what if V is evaluatively negative, e.g., *hate*?

- (45) I don't hate that Bo won the game.
 $\nrightarrow$ There exists someone x s.t. I would have
 hated it if x won the game

Evaluatively negative predicates (*hate*, *fear*) do not
 have a TSP: No triviality for *fear*!

Refining Uegaki & Sudo's generalization

An **evaluatively positive** non-veridical preferential predicate *V* **in general** cannot take questions

- ▶ **Exceptions:** *V whether S* means *V that S*.
- ▶ Restricted distribution, speaker disagreement, ...

Refining Uegaki & Sudo's generalization

An **evaluatively positive** non-veridical preferential predicate *V* **in general** cannot take questions

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An **evaluatively negative** non-veridical preferential predicate *V* is **more likely** to be able to take questions (because it does not have a TSP that leads to triviality)

- ▶ *V whether S* can mean *V that S* or *V that not S*
- ▶ But some might be anti-rogative for other independent reasons

Crosslinguistic predictions

Hope in general cannot take questions

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Crosslinguistic predictions

Hope in general cannot take questions

- ▶ Exceptions: *hope whether S* means *hope that S*.
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Fear is **more likely** to be able to take questions

- ▶ *fear whether S* can mean *fear that S* or *fear that not S*
- ▶ But *fear* might be anti-rogative for other independent reasons

Hope whether: crosslinguistic data

Turkish: *hope whether* $S \approx \textit{hope that } S$

- (46) John [komşusu evde olacak mı] diye
 John [his.neighbor home be.FUT Q C
 umuyor.
 hopes

“John hopes whether his neighbor will be home”

- ▶ True if John wants his neighbor to be home
- ▶ False if John doesn't want his neighbor to be home

Hope whether: crosslinguistic data

Mandarin Chinese: *hope whether S* $\approx$ *hope that S*

► Variation and limited distribution

- (47) % Yuehan xiwang Mali nengbuneng bangbang
 John hope Mary can.NEG.can help
 ta.
 him

“John hopes whether Mary can help him.”
 ($\approx$ John hopes that Mary can help him.)

Fear whether: crosslinguistic data

Besides Turkish *kork-* and Mandarin *haipa*, Japanese *osoreru* can take polar questions

- (49) Jon-wa [rinjin-ga shuumatsu ie-ni
 Jon-TOP [neighbor-NOM weekend home-LOC
 iru-ka] osoreteiru.
 stay-Q] fear

“John fears whether his neighbor will be home
 this weekend.”

- ▶ Compatible with John wanting his neighbor to be home (*fear that not S*)
- ▶ Compatible with John not wanting his neighbor to be home (*fear that S*)

Fear Q: further crosslinguistic data

In some languages, even though *fear* cannot take polar questions, it can take constituent questions.

Spanish *temer*:

- (50) * Juan teme si su vecino {estará /
 John fears if/whether his neighbor will.be /
 esté} en casa entonces.

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“John fears whether his neighbor will be home
 this weekend.”

- (51) Juan teme quién más pueda estar en
 John fears who more might.Pres.Subj be in
 el autobús.
 the bus

“John fears who else might be on the bus.”

Interim summary

Crosslinguistic evidence for a refined generalization

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Crosslinguistic evidence for a refined generalization

Hope in general cannot take questions

- ▶ Exceptions: *hope whether S* means *hope that S*.
- ▶ Restricted distribution, speaker disagreement, ...

Fear is more likely to be able to take questions

- ▶ *fear whether S* can mean *fear that S* or *fear that not S*
- ▶ But *fear* might be anti-rogative (or partially so) for other independent reasons

Conclusion

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- ▶ Theorists should definitely go beyond a handful of paradigmatic examples and test whether or to what extent their generalizations and formal accounts hold in the broader empirical landscape
- ▶ But interpreting the data is more complicated.
- ▶ Traditional, qualitative, in-depth analysis of individual items can complement standard statistical modeling/hypothesis testing

Conclusion

How should we use large-scale, quantitative experimental data to assess generalizations and formal analyses in semantics?

- ▶ Standard: testing certain summary statistics
- ▶ In-depth qualitative investigation on individual items is (also) important

How should we use corpus data/naturally occurring examples to assess generalizations and formal analyses in semantics?

- ▶ Avoid simplistic falsificationism
- ▶ Constructive theory building/revision

Conclusion

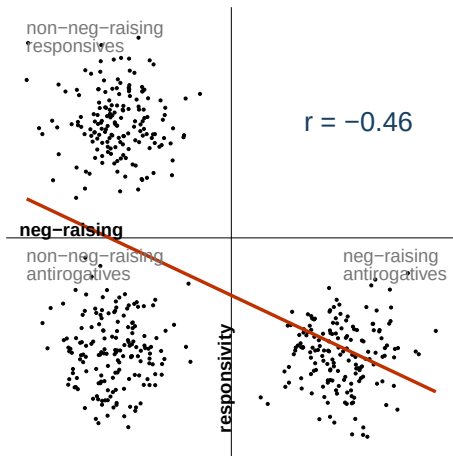
Crosslinguistic investigations are fruitful

- ▶ Many open empirical and theoretical issues

You can contribute!

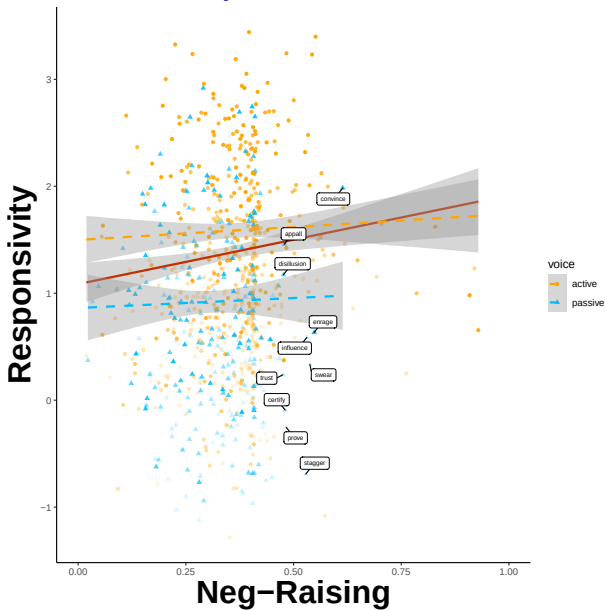
[https://wuegaki.ppls.ed.ac.uk/mecore/
mecore-databases/](https://wuegaki.ppls.ed.ac.uk/mecore/mecore-databases/)

Auxiliary distributional assumptions



Assumption: two classes of antirogatives with the same average responsivity scores

Correlations by Voice



Complication 2: Voice as a potential confound

- ▶ $r_{\text{overall}} = 0.11$ (95%CI = [0.03, 0.18])
- ▶ $r_{\text{active}} = 0.035$ (95%CI = [-0.06, 0.13])
- ▶ $r_{\text{passive}} = 0.027$ (95%CI = [-0.1, 0.16])

Alternative distributional assumptions:

- ▶ Predicates in active and passive voices each satisfy White's distributional assumptions ($r < 0.09$)
- ▶ Predicates in passive voices are (generally) not neg-raising and they have lower responsivity scores for reasons unrelated to (non-)neg-raising

The high overall correlation is explained away by Voice and need not falsify NR

Limitations of statistical testing

How should we use large-scale, quantitative experimental data to assess generalizations and formal analyses in semantics?

- ▶ Standard: testing certain summary statistics

Limitations:

- ▶ Statistical testing crucially relies on auxiliary distributional assumptions: different assumptions can lead to different results, but hard to know what the correct one would be
- ▶ Test result can only reject a hypothesis: no information about where and why it fails

White's counterexample 3

(52) I struggled to **believe whether** I could trust the Scriptures

► Is *believe* even neg-raising here?

Context: The speaker knows that John loved to go to bars in college but has not been in touch with him for a long time. When they met again years later, John told the speaker that he quit drinking shortly after graduation and had never been to a bar since. The speaker later comments about this:

(53) ??I struggled not to believe that John had been to a bar in years.

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